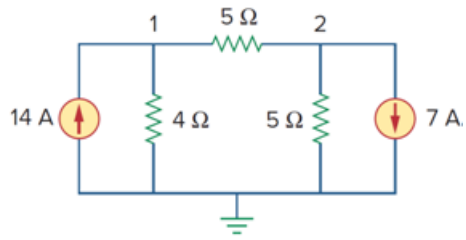


Problem

Obtain the node voltages in the circuit of Fig. 3.4.

Figure 3.4:



Step-by-step solution

Step 1 of 5

Refer to Figure 3.4 in the text book.

Identify the voltages in the circuit shown in Figure 1.

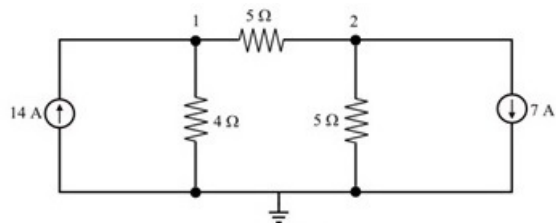


Figure 1

Step 2 of 5

The circuit is labeled as shown in Figure 2.

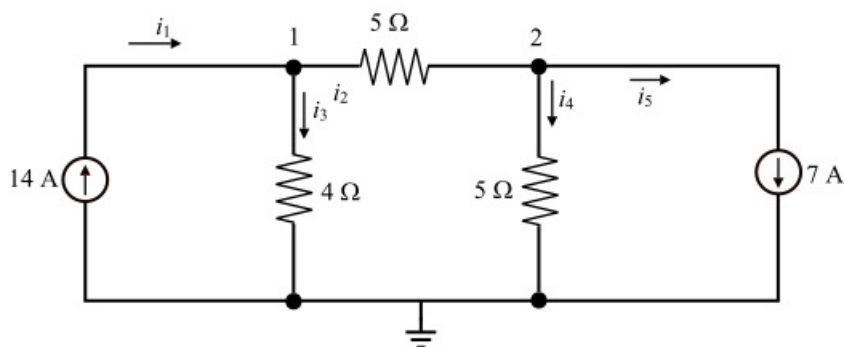


Figure 2

Step 3 of 5

Apply nodal analysis at node 1.

At node 1, applying Kirchhoff's current law, the current entering any node is equal to the current leaving from that node and Ohm's law gives

$$\frac{v_1}{4} - 14 + \frac{v_1 - v_2}{5} = 0$$

$$\frac{5v_1 - 280 + 4v_1 - 4v_2}{20} = 0$$

$$5v_1 - 280 + 4v_1 - 4v_2 = 0$$

$$9v_1 - 4v_2 - 280 = 0$$

$$9v_1 = 4v_2 + 280$$

$$v_1 = \frac{4v_2 + 280}{9} \dots\dots (1)$$

Therefore, the expression for the voltage v_1 is, $\frac{4v_2 + 280}{9}$.

Step 4 of 5

Apply nodal analysis at node 2.

$$\frac{v_2}{5} + 7 + \frac{v_2 - v_1}{5} = 0$$

$$\frac{v_2 + 35 + v_2 - v_1}{5} = 0$$

$$v_2 + 35 + v_2 - v_1 = 0$$

$$2v_2 - v_1 + 35 = 0$$

Substitute $\frac{4v_2 + 280}{9}$ for v_1 .

$$2v_2 - \left(\frac{4v_2 + 280}{9} \right) + 35 = 0$$

$$\frac{18v_2 - (4v_2 + 280) + 315}{9} = 0$$

$$18v_2 - (4v_2 + 280) + 315 = 0$$

$$18v_2 - 4v_2 - 280 + 315 = 0$$

$$14v_2 = -35$$

$$v_2 = -\frac{35}{14}$$

$$v_2 = -2.5 \text{ A}$$

Therefore, the value of the voltage v_2 is, -2.5 A.

Step 5 of 5

Recall equation (1).

$$v_1 = \frac{4v_2 + 280}{9}$$

Substitute -2.5 A for v_2 .

$$\begin{aligned} v_1 &= \frac{4(-2.5) + 280}{9} \\ &= \frac{-10 + 280}{9} \\ &= \frac{270}{9} \\ &= 30 \text{ A} \end{aligned}$$

Therefore, the value of the voltage v_1 is, 30 A.